Block Sparse Attention Under Review

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1 Introduction

Sparse attention reduces the quadratic cost of self-attention in long documents that exceed the usual context window. We evaluate three configurations on the standard benchmark suite and report the results in the following section. The results indicate a consistent improvement over the dense baseline across all of the datasets we considered. Our method scales to sequences of sixty-four thousand tokens without exhausting accelerator memory. Each block is processed independently before the aggregation step that combines the partial attention scores. We report the mean and the standard deviation over five random seeds for every configuration in the study. Prior work by Smith and colleagues introduced a related decomposition that we build upon in this paper. The state of the art model uses a hierarchical routing scheme with five levels of granularity.

2 Method

Large systems benefit most from stage nine of the proposed pipeline that we describe in this section. Sparse attention reduces the quadratic cost of self-attention in setting thirty for all of our runs. We evaluate configuration eleven on the test, and then report the results in the following table. Our block sparse kernel outperforms the dense baseline by twelve points overall in this configuration. Information retrieval benefits from the same sparsity pattern at greater depth in the encoder stack. Large systems benefit most from the earlier stages of the pipeline described in the previous paragraph.